CONVERGENCE AND ERROR ANALYSIS OF REGULA FALSI METHOD

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INTRODUCTION (WHAT, ABOUT, WHY)

What is it?

- Also known as the Method of False Position.
- It is a numerical method for solving an equation in one unknown. It is a "bracketing method," meaning it requires two initial guesses that bracket the root.

About the Logic:

- It combines features of the Bisection Method (bracketing reliability) and the Secant Method (linear interpolation speed).
- It assumes the function is linear in the interval and calculates the x-intercept of the line connecting the two points.

Why use it?

- Guaranteed Convergence: Unlike the Newton-Raphson method, Regula Falsi is guaranteed to converge as long as the initial guesses bracket the root.
- Simplicity: It does not require calculating the derivative of the function.

DERIVATION OF THE METHOD

Geometric Concept:

- Consider a curve $y = f(x)$. We choose two points a and b such that $f(a)$ and $f(b)$ have opposite signs (i.e., $f(a) \cdot f(b) < 0$).
- We draw a chord (secant line) connecting $(a, f(a))$ and $(b, f(b))$. The point where this chord intersects the x-axis is our next approximation, c.

Equation of the Chord:

- Using the two-point form:

$$y - f(a) = \frac{f(b) - f(a)}{b - a}(x - a)$$

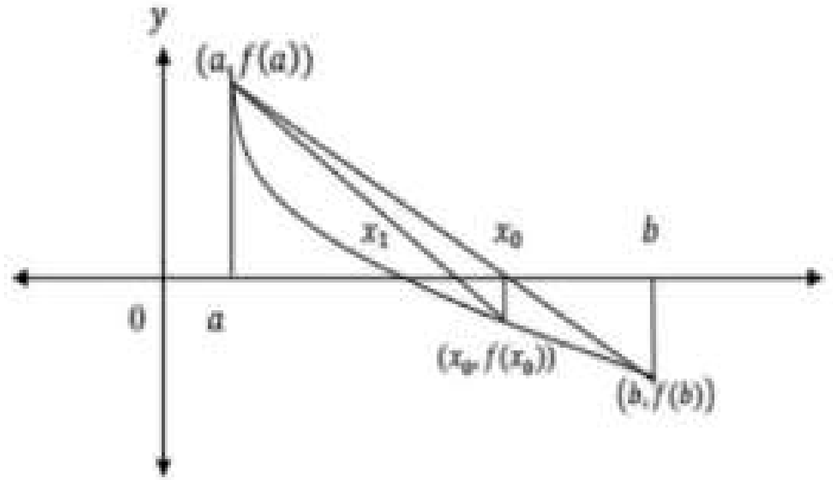
Finding the Root (c):

- At the x-intercept, $y = 0$ and $x = c$. Substituting these values:

$$c = a - \frac{f(a) \cdot (b - a)}{f(b) - f(a)}$$

- Simplifying, we get the standard Regula Falsi formula:

$$c = \frac{af(b) - bf(a)}{f(b) - f(a)}$$



ERROR & CONVERGENCE ANALYSIS

- Convergence Analysis:
- Order of Convergence: The order of convergence is 1 (Linear).
- This means the error decreases linearly with each iteration.
- While slower than the Newton-Raphson method (Quadratic convergence), it is faster than the Bisection method.
- Error Analysis:
- The method can suffer from "stagnation" if the curve is highly convex or concave. One endpoint of the interval may remain fixed, leading to slower convergence towards the end.
- **Stopping Criteria:** The algorithm stops when the absolute error $|x_{new} - x_{old}| < \epsilon$ (tolerance) or $|f(c)| \approx 0$.

ENGINEERING APPLICATION/ EXAMPLE

Find the root of the equation $f(x) = x^3 - x - 2 = 0$ using the **Regula Falsi method**, and study its **error and convergence behavior.**

Evaluate:

- $f(1) = -2$
- $f(2) = 4$

Since:

$$f(1)\,f(2) < 0$$

A root exists in $[1, 2]$.

Regula Falsi formula:

$$x_n = \frac{a_n f(b_n) - b_n f(a_n)}{f(b_n) - f(a_n)}$$

Iteration	x_n	Absolute Error $\mid e_n \mid = \mid x_n - \alpha \mid$
1	1.3333	0.1881
2	1.4627	0.0587
3	1.5040	0.0174
4	1.5163	0.0051
5	1.5199	0.0015

Error Analysis

$$\frac{\mid e_2 \mid}{\mid e_1 \mid} \approx \frac{0.0587}{0.1881} \approx 0.31$$

$$\frac{\mid e_3 \mid}{\mid e_2 \mid} \approx 0.30$$

$$\frac{\mid e_4 \mid}{\mid e_3 \mid} \approx 0.29$$

✓The ratio is approximately **constant**

✓ Error reduces **proportionally**

$$\boxed{\mid e_{n+1} \mid \approx C \mid e_n \mid \ (0 < C < 1)}$$

Convergence Analysis

A method converges with order $p$if:

$$\lim_{n \to \infty} \frac{\mid e_{n+1} \mid}{\mid e_n \mid^p} = \lambda$$

For Regula Falsi

$$\frac{\mid e_{n+1} \mid}{\mid e_n \mid} \to \lambda \neq 0$$

Hence:

$$\boxed{p = 1}$$

→Regula Falsi is linearly convergent

REFERENCES

Books:

- Numerical Methods in Engineering & Science by B.S. Grewal.
- Introductory Methods of Numerical Analysis by S.S. Sastry.

Journal:

- SIAM Journal on Numerical Analysis - Articles on bracketing methods and optimization.

Online Portals:

- NPTEL (IIT Kharagpur): Course on "Numerical Methods for Engineers."
- Wolfram MathWorld: Validated entries on "Method of False Position."

THANK YOU